

FBS Project: What the Three Tasks Involve

Notes on tasks 1 to 3

Task 1: Implement fbs.py for one time step

For this task fbs.py only has to solve the network at one moment. We give it the network data (the line impedances, the load at each bus and the PV output at that moment) and it gives back the bus voltages, the line currents and the losses for that moment. It doesn't look at the hour before or the hour after.

We kept it to one time step on purpose. If the solver also had to loop over a whole day and decide what the battery does, everything would be mixed together and it would be hard to tell where a mistake came from. This way fbs.py does one job. A separate script can then call it once for each hour, changing the load and PV values each time, and save the results. That is how run_simulation.py works at the moment, and fbs.py didn't need any changes for it.

Task 2: Verify the FBS result

Getting a result without an error doesn't mean the result is right. A power flow program can give voltages around 0.97 or 0.98 that look fine and still have a wrong sign or a mistake in how the buses are connected. If that goes unnoticed, every later result built on it is wrong too, including the battery sizing.

We used a few checks. First we solved a small 3-bus network by hand with a calculator, then ran the same network through fbs.py. Both gave the same voltages, the same angles and the same losses.

Second, we checked the power balance on the 5-bus case. The power coming out of the substation should equal the total load plus the line losses, and it does: 733.963 kW from the substation, against 720.000 kW of load plus 13.963 kW of losses. This check doesn't depend on how the code is written, so it can catch mistakes like counting a load twice or leaving out the loss of a line.

Third, we made a rough estimate of the voltage at bus 2 using the voltage drop formula, and it was close to what the solver gave (0.9747 against 0.9748).

None of these is enough on its own. The hand calculation shows the algorithm is coded correctly, and the power balance shows the totals add up on the actual 5-bus network.

Task 3: Show basic network results

This task is about showing the results so people can read them. The solver gives complex numbers for the voltages and currents, which aren't easy to look at, so we turn them into what we actually want to know: the voltage at each bus compared to the substation, the current and power in each line, and the total losses.

We used tables and a chart. The tables give exact numbers that we can quote in the report or compare with the hand calculation. The chart shows the voltage profile. On a radial feeder it should go down steadily as you move away from the substation, because each line drops the voltage a bit more. If the chart went up somewhere or jumped around, we would know something was wrong before reading any single number. So this step is also a last check on the whole thing. In our case bus 4 has the lowest voltage, which makes sense because it is the furthest bus from the substation.

The code, the check calculations and the results themselves are in the implementation report (Tasks_12-19_FBS_Implementation_Report.pdf).